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Solution-Operator Semi-Discretization: General-Order Multiplication-Free Stability Analysis of Time-Periodic Delayed Systems and the Accuracy–Sparsity Trade-off

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Abstract

The multiplication-free semi-discretization method (MFSD) reduces the stability analysis of time-periodic delay differential equations to a sparse, banded generalized eigenvalue problem with linear time complexity in the resolution p; however, the classical piecewise-constant step limits its convergence order to two. This paper introduces the *solution-operator semi-discretization* (SOSD) method: since the multiplication-free representation of the one-period solution operator only requires a per-step residual, the integrator inside each step becomes a free design choice. The residual block rows of any explicit Runge–Kutta or implicit collocation-type scheme (Gauss–Legendre, Radau, Lobatto) are embedded in the banded system through a Kronecker-product structure of the Butcher tableau and the system matrices, raising the convergence order of the dominant characteristic multipliers to the order of the chosen scheme — super-convergent order $2s$ for s-stage Gauss quadrature points — provided the delayed terms are interpolated by the continuous extension of the same scheme. In the limit of a single ultra-high-order step, the framework degenerates to the known global spectral method; between the two extremes lies a continuum in which per-step order is traded against matrix sparsity. Because the number of nonzeros grows cubically with the stage number while the error decreases only exponentially, the CPU-time optimum for a prescribed accuracy is a problem-dependent *sweet spot* at moderate order and many steps: its location is measured for systems requiring only tens of points per period and for systems — high natural frequency relative to the period — requiring many hundreds, where the banded moderate-order route wins decisively. We further argue that previous method comparisons conflated p-refinement with h-refinement, and we prescribe a fair work–precision methodology (log–log axes, wide resolution ranges, variance-reported CPU times, one machine and thread configuration). All methods are released in the open-source Julia package SOSD.jl.

Keywords

Time-delay systems, semi-discretization, solution operator, Runge–Kutta methods, Floquet theory, stability analysis, computational complexity

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Introduction

Time-periodic linear delay differential equations (DDEs) of the form

$$\dot{\mathbf{x}}(t) = \mathbf{A}(t)\,\mathbf{x}(t) + \sum_{k=1}^{g} \mathbf{B}_k(t)\,\mathbf{x}\big(t - \tau_k(t)\big) + \mathbf{c}(t), \tag{1}$$

with T-periodic coefficients $\mathbf{A}(t)$, $\mathbf{B}_k(t)$, delays $\tau_k(t)$ and forcing $\mathbf{c}(t)$, govern a wide range of engineering and biological systems: regenerative machine-tool chatter[1,13,17], turning with spindle-speed variation[12,15], feedback control with latency[16], and seasonally forced population dynamics. Their asymptotic stability is decided by the dominant characteristic multipliers of the monodromy operator associated with one period[10].

Two families of numerical methods dominate practice. Semi-discretization (SD)[9,10] advances the state with a fixed step $\Delta t = T/p$, treating the delayed term (and, in the classical zeroth- and first-order variants, the coefficients) as piecewise constant or piecewise linear per step; its convergence order is limited to two[11]. Collocation and pseudospectral methods[4,5] represent the solution segment by a global high-order polynomial and converge spectrally for smooth problems. A recurring claim in the literature is that collocation "outperforms" semi-discretization; we argue in Section that such comparisons are structurally unfair, because they compare a p-refinement strategy (raising the polynomial order) with an h-refinement strategy (shrinking the step) — the same category error as declaring p-type finite elements universally superior to h-type ones.

The recent multiplication-free semi-discretization method (MFSD)[2] removed the main cost bottleneck of SD: instead of multiplying p one-step matrices into a dense monodromy matrix ($\mathcal{O}(p^{2.8})$ in practice), every step is written as a residual equation and all p steps are stacked into one sparse banded system, whose one-period transition is obtained from the generalized eigenvalue problem $\mathbf{\Phi}_{\mathrm{L}}\mathbf{y}_p = \mathbf{\Phi}_{\mathrm{R}}\mathbf{y}_0$ at $\mathcal{O}(p)$ cost. The MFSD assembly is inherited machinery here and is *not* claimed as a contribution of the present paper.

The present paper starts from a simple observation: *the multiplication-free form only needs a per-step residual.* Semi-discretization is, structurally, a fixed-step time integrator whose only approximation is the per-step treatment of the coefficients and the delayed term. Consequently, the piecewise-constant step can be replaced by *any* one-step scheme — an explicit Runge–Kutta method or an implicit collocation-type method — by placing the Butcher-tableau coefficients into the residual block rows and augmenting the per-step state with the internal stage values. This raises the achievable convergence order arbitrarily, at a quantifiable structural price: the sparse matrices become denser as the stage number grows. We call the resulting family the *solution-operator semi-discretization* (SOSD): the banded pair $(\mathbf{\Phi}_{\mathrm{L}}, \mathbf{\Phi}_{\mathrm{R}})$ is a sparse representation of the one-period solution operator of (1), and the per-step integrator merely selects the order of that representation.

The contributions are, honestly delimited:

1. the systematic embedding of arbitrary Runge–Kutta and collocation steps (via their Butcher tableaux and continuous extensions) into the multiplication-free banded stability framework for time-periodic DDEs, including time-periodic delays (Section);
2. the explicit-versus-implicit matrix-structure and CPU trade-off analysis, and the demonstration that the single-step limit reproduces the known global collocation method (Section);
3. the identification and characterization of the accuracy–sparsity *sweet spot*: for a given accuracy target the CPU optimum lies at moderate per-step order and many steps, at a location that depends on the problem dimension and resolution demand (Section);
4. a fair-comparison methodology that separates p- and h-refinement and reports work–precision curves with variance on one fixed machine configuration (Section); and
5. an open-source Julia implementation, SOSD.jl, with all scripts needed to reproduce every figure.

Neither collocation itself[4] nor the multiplication-free assembly[2] is claimed as new; the value of the paper is the unifying embedding, the trade-off quantification, and the practical guidance it yields.

Background: MFSD and semi-discretization as fixed-step integration

Discretize one principal period into p uniform steps, $\Delta t = T/p$, $t_n = n\,\Delta t$, and let $r = \lceil \tau_{\max}/\Delta t \rceil$ count the steps covering the largest delay. Classical SD constructs a one-step mapping $\mathbf{x}_{n+1} = \mathbf{P}_n\mathbf{x}_n + \mathbf{R}_n\mathbf{x}_{n-r}$ from the exact solution of the frozen-coefficient system over $[t_n, t_{n+1}]$[10]. The monodromy matrix is the product of p such mappings acting on the discretized history $\mathbf{y}_n = (\mathbf{x}_n^\top, \ldots, \mathbf{x}_{n-r}^\top)^\top$; forming this product costs $\mathcal{O}(p\,(rd)^\omega/r^\omega)$-ish work in practice ($\approx \mathcal{O}(p^{2.8})$ measured in[2]), and its avoidance is the point of MFSD: each step contributes the residual row $\mathbf{0} = \mathbf{I}\mathbf{x}_{n+1} - \mathbf{P}_n\mathbf{x}_n - \mathbf{R}_n\mathbf{x}_{n-r}$ to a sparse banded block system over the extended vector spanning $[t - \tau_{\max},\, t + T]$, split into left/right blocks so that the spectrum follows from $\mathbf{\Phi}_{\mathrm{L}}\mathbf{y}_p = \mu\,\mathbf{\Phi}_{\mathrm{R}}\mathbf{y}_0$ via sparse Krylov iteration[7,14] at $\mathcal{O}(p)$ total cost[2].

Two properties of this construction matter for what follows. First, the one-step map is a *fixed-step integrator*: the only approximation is that $\mathbf{A}(t)$, $\mathbf{B}(t)$ and the delayed argument are frozen (or low-order interpolated) across Δt. It is exactly this per-step treatment that caps the convergence order at two: as shown by Insperger et al.[11], a first-order approximation of the delayed term is already optimal for the piecewise-constant coefficient scheme, and no refinement of the classical SD step exceeds order two. Second, the assembly touches the integrator only through the per-step residual; nothing in the banded structure, the left/right splitting for $p-1 \gtrless r$, or the Krylov solver depends on *which* integrator produced the residual. The integrator is a free design choice — the theme of this paper.

Higher-order multiplication-free stepping

Stage-augmented residual blocks from a Butcher tableau

Consider an s-stage Runge–Kutta method with tableau $(\mathbf{a}, \mathbf{b}, \mathbf{c})$, a_{ij}, b_j, c_j, applied to (1) on $[t_n, t_n + \Delta t]$. Writing the stage *values* $\mathbf{Y}_i \approx \mathbf{x}(t_n + c_i\Delta t)$ (rather than stage slopes), the step reads

$$\mathbf{Y}_i = \mathbf{x}_n + \Delta t \sum_{j=1}^{s} a_{ij} \Big[\mathbf{A}(t_{nj})\,\mathbf{Y}_j + \mathbf{d}_j\Big], \qquad i = 1, \ldots, s, \tag{2}$$

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \Delta t \sum_{j=1}^{s} b_j \Big[\mathbf{A}(t_{nj})\,\mathbf{Y}_j + \mathbf{d}_j\Big], \tag{3}$$

with $t_{nj} = t_n + c_j\Delta t$ and the delayed-plus-forcing terms

$$\mathbf{d}_j = \sum_{k=1}^{g} \mathbf{B}_k(t_{nj})\,\tilde{\mathbf{x}}\big(t_{nj} - \tau_k(t_{nj})\big) + \mathbf{c}(t_{nj}), \tag{4}$$

where $\tilde{\mathbf{x}}$ denotes the interpolated delayed state (Section). The per-step unknown block is the stage-augmented vector

$$\mathbf{v}_n = \big(\mathbf{x}_n^\top,\, \mathbf{Y}_1^\top, \ldots, \mathbf{Y}_s^\top\big)^\top \in \mathbb{R}^{(s+1)d}, \tag{5}$$

and the global unknown $\mathbf{q} = (\mathbf{v}_1^\top, \ldots, \mathbf{v}_p^\top)^\top$ collects all steps of one period. Equations (2)–(3) are linear in the current and delayed states, so each step contributes one block row to a sparse system $\mathbf{\Phi}_{\mathrm{L}}\,\mathbf{q} = \mathbf{\Phi}_{\mathrm{R}}\,\mathbf{q}_{\mathrm{hist}}$, exactly as in MFSD but with $(s+1)d$-sized blocks.

The structure of these blocks is a Kronecker (dyadic) composition of three independent ingredients: the Butcher tableau, the system matrices, and the interpolation weights. Collecting the tableau into the compound $(s{+}1) \times s$ matrix

$$\hat{\mathbf{A}} = \begin{bmatrix} \mathbf{a} \\ \mathbf{b}^\top \end{bmatrix}, \qquad \boldsymbol{\mathcal{A}}_n = \mathrm{diag}\big(\mathbf{A}(t_{n1}), \ldots, \mathbf{A}(t_{ns})\big), \tag{6}$$

the step (2)–(3) reads, for the stacked unknowns $\mathbf{w}_n = (\mathbf{Y}_1^\top, \ldots, \mathbf{Y}_s^\top, \mathbf{x}_{n+1}^\top)^\top$,

$$\Big[\mathbf{I} - \Delta t\,\big(\hat{\mathbf{A}} \otimes \mathbf{I}_d\big)\,\boldsymbol{\mathcal{A}}_n\,\mathbf{S}\Big]\mathbf{w}_n = \big(\mathbf{1}_{s+1} \otimes \mathbf{I}_d\big)\,\mathbf{x}_n + \Delta t\,\big(\hat{\mathbf{A}} \otimes \mathbf{I}_d\big)\,\mathbf{d}_n, \tag{7}$$

where $\mathbf{S} = [\,\mathbf{I}_{sd}\ \ \mathbf{0}\,]$ selects the stage part of $\mathbf{w}_n$ and $\mathbf{d}_n$ stacks the terms (4). Each delayed sample in $\mathbf{d}_n$ is itself a dyad: stage j of delay k contributes

$$\Delta t\,\big(\hat{\mathbf{A}}\,\mathbf{e}_j \otimes \mathbf{B}_k(t_{nj})\big)\,\big(\mathbf{w}(\theta_{njk})^\top \otimes \mathbf{I}_d\big), \tag{8}$$

i.e. the outer product of a Butcher-tableau column, the delay coefficient matrix, and the continuous-extension weight row $\mathbf{w}(\theta)^\top$ evaluated at the fractional position θ_{njk} of the delayed instant. The entire assembly therefore separates into precomputable factors (tableau, weights) and pointwise coefficient evaluations

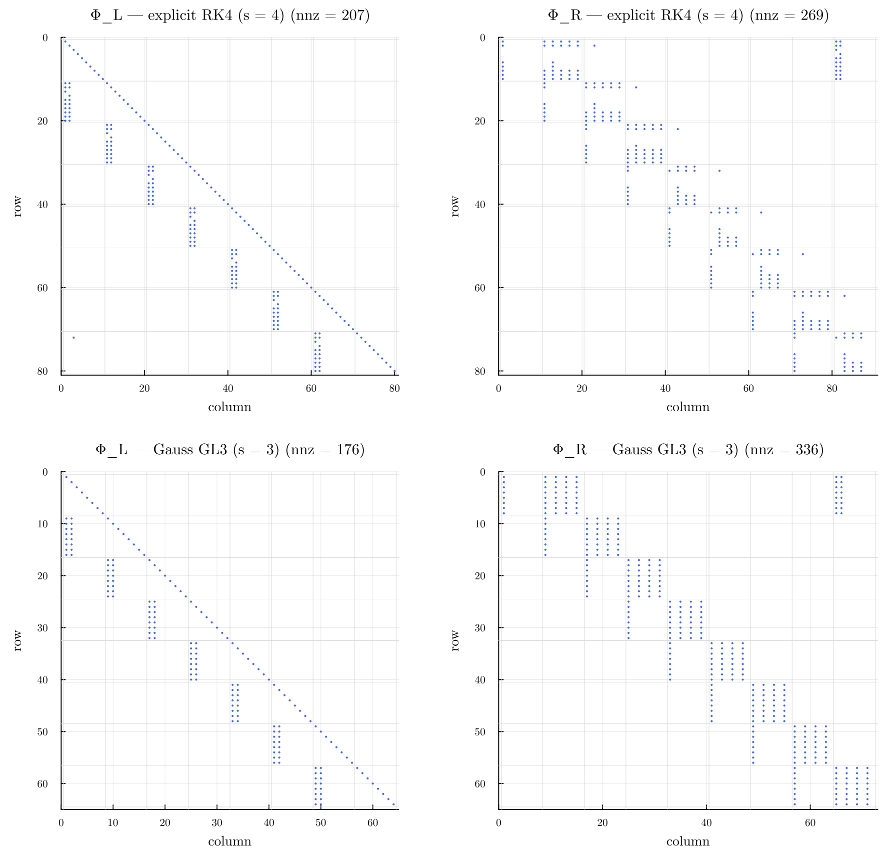

Figure 1. Sparsity patterns of the solution-operator pair $(\mathbf{\Phi}_{\mathrm{L}}, \mathbf{\Phi}_{\mathrm{R}})$ on the delayed Mathieu equation ($p = r = 8$): explicit RK4 (top) versus Gauss GL3 (bottom). $\mathbf{\Phi}_{\mathrm{L}}$ is unit block-lower-triangular in both cases (forward-substitution solvable, no fill-in); the delay band sits in $\mathbf{\Phi}_{\mathrm{R}}$, with denser per-block coupling for the collocation-type stages.

$\mathbf{A}(t^*)$, $\mathbf{B}_k(t^*)$ — no matrix products between steps ever occur. Figure 1 shows the resulting sparsity patterns: $\mathbf{\Phi}_{\mathrm{L}}$ has unit diagonal and is strictly block-lower triangular for *both* explicit and implicit tableaux (after the per-step elimination below), so linear solves reduce to forward substitution with zero fill-in; all delay couplings that reach into the pre-period history populate $\mathbf{\Phi}_{\mathrm{R}}$. The per-block fill of the delay band grows with the stage number — the mechanism that will drive the accuracy–sparsity sweet spot.

In the implementation the stage equations (2) are *pre-eliminated* per step: the stage-coupling matrix $\mathbf{M}_n = \mathbf{I} - \Delta t\,(\mathbf{a} \otimes \mathbf{I}_d)\,\mathrm{diag}\big(\mathbf{A}(t_{n1}), \ldots, \mathbf{A}(t_{ns})\big)$ of size $sd \times sd$ is factorized once (LU; a dense inverse for small sd), and the responses of $(\mathbf{x}_{n+1}, \mathbf{Y}_1, \ldots, \mathbf{Y}_s)$ to (i) $\mathbf{x}_n$, (ii) each delayed-state sample, and (iii) the additive forcing are precomputed as $(s+1)d \times d$ transition blocks. This keeps

$\mathbf{\Phi}_{\mathrm{L}}$ *unit block-lower triangular* — its LU factorization is trivial and fill-free — while the per-step elimination costs $\mathcal{O}\big(p\,(sd)^3\big)$, linear in p. For explicit tableaux (strictly lower-triangular $\mathbf{a}$) the elimination degenerates to forward substitution; for implicit collocation tableaux (Gauss–Legendre of order $2s$, Radau IIA of order $2s-1$, Lobatto IIIA of order $2s-2$ [6,8]) it is a genuine dense solve, which is precisely where the super-convergent order is purchased.

Delay interpolation by continuous extension

The delayed argument $t_{nj} - \tau_k(t_{nj})$ generally falls between grid nodes, inside some earlier step m. Interpolating the delayed state with an order lower than the integrator silently caps the observed convergence order — the higher-order analogue of the remark of Insperger et al. [11] for the classical scheme, and a classical result of DDE numerics [3]. The consistent choice is the *continuous extension* of the step itself: for collocation methods, the collocation polynomial through $\big(\mathbf{x}_m, \mathbf{Y}_{1,m}, \ldots, \mathbf{Y}_{s,m}, \mathbf{x}_{m+1}\big)$ evaluated at the fractional position $\theta \in [0,1]$ of the delayed instant inside step m; for explicit methods, the scheme's dense-output polynomial (for RK4 the classical cubic Hermite-type formula). The interpolation weights depend only on θ and the tableau, so they join the precomputed transition blocks without changing the banded structure; each delayed sample couples block row n to the two neighbouring history blocks m and $m+1$. Since the delayed state lies at least one full delay in the past, these couplings stay strictly below the diagonal and $\mathbf{\Phi}_{\mathrm{L}}$ remains triangular; couplings reaching into the pre-period history enter $\mathbf{\Phi}_{\mathrm{R}}$ instead. Time-periodic delays $\tau_k(t)$ are handled naturally, because θ and the target block m are evaluated per stage from $\tau_k(t_{nj})$.

The single-step limit is global collocation

Setting $p = 1$ with an s-stage Gauss tableau makes the single step span the entire period: the stage values *are* then the collocation samples of a global polynomial, and the construction reproduces the known collocation/pseudospectral method for DDEs [4,5] in its strong form. This extreme is explicitly *not* new; we include it (i) to make the equivalence concrete — in Section the single-step spectrum matches the converged reference to near machine precision — and (ii) to expose its cost: the single-step matrix is effectively dense, and the number of nonzeros of the assembled system grows as $\mathcal{O}(d^2 s^3)$ with the stage count, so pushing all accuracy into one step abandons exactly the sparsity that makes the multiplication-free form fast. The continuum between "p large, s small" and "$p = 1$, s large" is the design space this paper maps.

Fair-comparison methodology

Comparisons between DDE stability methods frequently pit a p-refinement method (collocation at increasing polynomial order) against an h-refinement method (semi-discretization at decreasing step size) and report accuracy at a single, often low, resolution on linear axes. Such comparisons cannot identify asymptotic rates, hide constant factors, and — like comparing p-type against h-type FEM refinement — do not license a universal winner. The predecessor paper[2] already criticized sub-100-resolution, non-log-log comparisons; we extend that critique into a positive prescription:

1. **Work–precision axes.** The primary comparison is CPU time versus eigenvalue error, both logarithmic, over wide ranges; the secondary one is CPU time versus resolution p with fitted scaling exponents.
2. **Variance-reported timing.** Every timed point is a warm-started mean over repeated runs with its standard deviation.
3. **Both refinement families on the same axes.** Every method is shown under h-refinement, and the per-step order s is swept separately (Section) so the full (p, s)-surface is visible, not a single slice.
4. **Fixed environment.** One machine, one BLAS thread configuration, pinned package versions, deterministic start vectors; all reported alongside the results.
5. **Verified references.** Error is measured against a reference that two distinct resolutions of a higher-order method agree on, and the reference method is cross-validated against an independent implementation (SemiDiscretizationMethod.jl).

Numerical results

[This section is populated from benchmark/results/*.csv; numbers below marked in red are placeholders until the full suite finishes.]

Test systems

Four systems span the design space (parameters in Appendix):

1. **Delayed Mathieu equation** ($d = 2$, $\tau = T = 2\pi$): the canonical smooth validation case[10].
2. **Seasonal scalar model** ($d = 1$, $\tau = 2$, $T = 2\pi$): a smooth, cheap system where high per-step order shines.
3. **Turning with spindle-speed variation** ($d = 2$, time-periodic delay, $T/\tau \approx 10$): exercises the $p \gg r$ partitioning and time-varying delay handling.
4. **FEM beam with delayed boundary feedback** ($d = 28$, $\tau/T = 1/2$), optionally with act-and-wait switching of the feedback (non-smooth stress test): the high-dimensional, high-resolution case where banded moderate order should win.

Order verification

Figure 2 and Table 1 confirm on the delayed Mathieu equation that each embedded *collocation* integrator attains its theoretical convergence order in the dominant multiplier: the Gauss family exhibits the super-convergent order $2s$ (fitted slopes 2.00, 4.00, 5.99 and 9.96 for $s = 1, 2, 3, 5$; GL8 reaches the 10^{-16}

Table 1. Order verification on the delayed Mathieu equation: nominal versus fitted convergence order of the dominant multiplier (log–log slope over the pre-floor error range).

Method	Nominal	Fitted	Comment
SDM (order 2)	2	3.1	delayed-term interpolation superconverges
RK2 (Heun)	2	2.0	
RK4	4	3.3	capped by 3rd-order dense output
RK5	5	2.1	capped by generic stage interpolant
GL1 (midpoint)	2	2.0	
GL2	4	4.0	superconvergence $2s$
GL3	6	6.0	
GL5	10	10.0	
GL8	16	—	at 10^{-16} floor from $p = 16$
Radau IIA, s=3	5	4.9	
Lobatto IIIA, s=3	4	4.0	
Lobatto IIIA, s=4	6	6.0	

relative floor already at $p = 16$), Radau IIA attains $2s - 1$ and Lobatto IIIA $2s - 2$. The explicit family behaves differently, and instructively: the observed order is the *minimum* of the method order and the order of its continuous extension. RK4 with its classical third-order dense output converges with slope ≈ 3.3 instead of 4, and RK5 with a generic stage-value interpolant collapses to slope ≈ 2.1 — the delayed lookup, not the step, is the binding constraint, exactly as anticipated in Section . For collocation tableaux this constraint is vacuous because the stage values lie on the collocation polynomial, whose degree grows with s; the matching-order continuous extension is therefore available *for free*. This asymmetry is a structural argument for implicit collocation steps inside the multiplication-free framework that the usual "explicit steps are cheaper" intuition misses.

Work–precision and time complexity

Figure 3 shows the three canonical panels for the delayed Mathieu equation: error versus p, CPU time versus p with $\pm 1\sigma$ bands, and the work–precision plane. Every embedded integrator preserves the linear time complexity of the multiplication-free assembly — the fitted scaling exponents are 1.0 for all methods, from RK1 through GL8 — so raising the per-step order shifts the CPU-time line by a constant factor (the block fill) without changing its slope. On the work–precision plane this is decisive: GL5 and GL8 reach relative errors of 10^{-14} within milliseconds, whereas the order-2 baseline spends an order of magnitude more time to reach only 10^{-6}. The same qualitative picture holds for the scalar seasonal model and for the spindle-speed-variation turning system (time-periodic delay, $p \gg r$ partitioning); the beam is discussed with the sweet-spot study below. [beam WP numbers after suite completes]

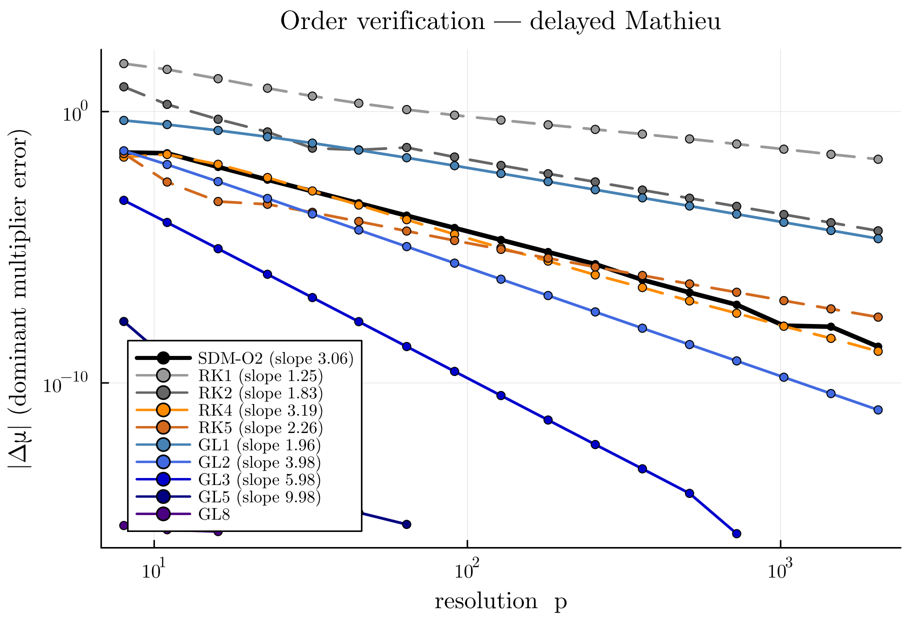

Figure 2. Order verification on the delayed Mathieu equation: eigenvalue error versus resolution p (log–log), one curve per method, dashed lines showing fitted slopes. [regenerate from results/order_verification_mathieu.csv]

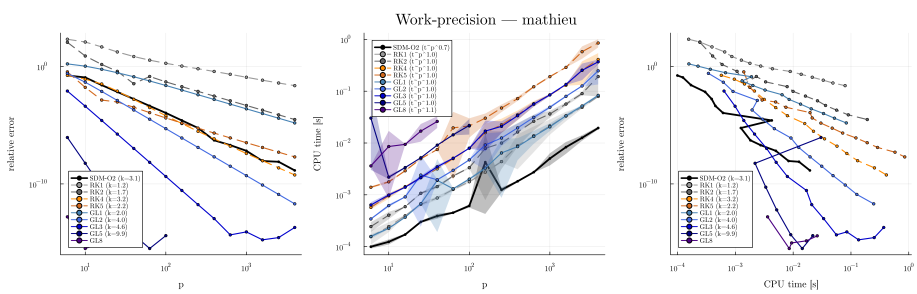

Figure 3. Work-precision study on the delayed Mathieu equation. Left: eigenvalue error vs. resolution p; middle: CPU time vs. p (mean of repeated runs, shaded $\pm 1\sigma$), all fitted exponents ≈ 1.0; right: the work–precision plane (down and left is better).

For baseline parity with the predecessor [2], Figure 4 reproduces, in the present environment, the time-complexity contrast between the traditional monodromy construction (SD-classic, fitted exponent 2.7, in line with the $\approx$2.8 reported in [2]) and the banded LR mapping (fitted exponent 1.1); both agree in the multiplier to 10^{-15}, and at $p = 1778$ the multiplication-free route is already 2500$\times$ faster.

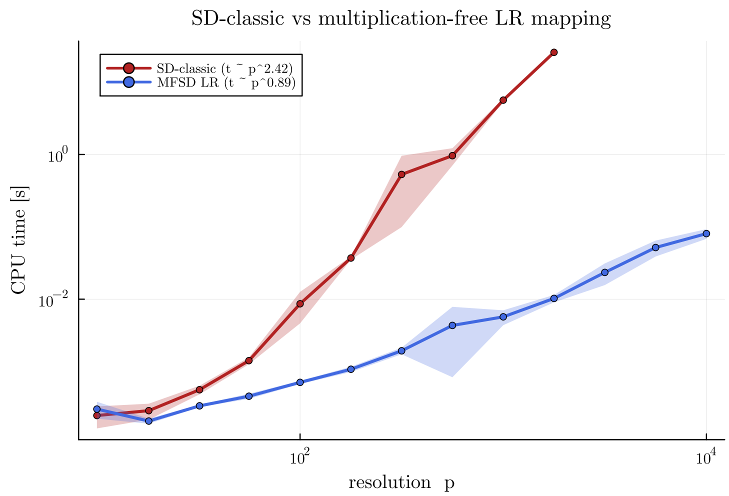

Figure 4. Baseline parity with the predecessor: CPU time of the traditional monodromy build (SD-classic) versus the banded multiplication-free mapping on the delayed Mathieu equation, mean $\pm 1\sigma$, with fitted scaling exponents.

The accuracy–sparsity sweet spot

For a fixed relative accuracy target ε, sweeping the per-step order s while choosing the smallest resolution p that reaches ε produces a U-shaped CPU-time curve: low orders need excessive p, high orders pay cubically growing per-step fill, $\mathrm{nnz}(\boldsymbol{\Phi}_{\mathrm{L}}) \sim p\, d^2 s^3$. A first-order cost model, cost $\propto p\, s^3$ with $\varepsilon \propto (T/p)^{2s}$, predicts the optimum

$$s^* \approx \tfrac{1}{6} \ln(1/\varepsilon), \tag{9}$$

i.e. even a 10^{-12} accuracy target asks for only $s^* \approx 5$ Gauss stages — moderate order, many steps. The measured optima ([Mathieu: $s^* = \ldots$; beam: $s^* = \ldots$ at $\varepsilon = 10^{-5}, 10^{-8}, 10^{-11}$]) confirm both the U-shape and its logarithmic drift with ε, and the beam shifts the optimum to lower order than the small Mathieu system, as predicted by the d^2-scaling of the fill.

Contrasting resolution demands: where the banded route wins

The location and the *stakes* of the sweet spot depend on how many points per period the problem physically requires. Two showcases bracket the range:

- **Low demand (tens of points).** For the delayed Mathieu equation and the seasonal scalar model, $p \approx 10$–30 already reaches deep accuracy with moderate Gauss orders; matrices are so small that sparsity is irrelevant, and even the single-step spectral corner is viable.
- **High demand (hundreds of points and more).** The spindle-speed-variation turning system oscillates with $T/T_{\mathrm{osc}} \approx 36$ (natural frequency high relative to the period), so *any* method needs $p \gtrsim 500$ to resolve

the dynamics at all. Here the banded moderate-order route wins decisively: the fill of a high-order or single-step variant is prohibitive at such p, while the $\mathcal{O}(p)$ banded solve is unaffected. [turning sweet-spot numbers: optimal (s^*, p^*) and CPU vs high-order alternative, from sweet_spot_turning_ssv.csv]

The same contrast governs high-dimensional systems: for the FEM beam ($d = 28$), memory — the $d^2 s^3$ per-step fill — caps the affordable stage number before CPU time does.

Explicit versus implicit steps

The conventional trade-off — explicit steps are cheap but order-capped, implicit stages are expensive but super-convergent — is reshaped by two findings. First, after per-step pre-elimination *both* families yield a unit block-lower-triangular $\mathbf{\Phi}_{\mathrm{L}}$, so the downstream solve cost differs only through the block fill; the explicit advantage reduces to the cheaper (forward-substitution) per-step elimination, which is a linear-in-p term either way. Second, the delay-interpolation order requirement (Table 1) penalizes exactly the explicit family: sustaining order $q > 3$ explicitly requires bespoke dense output of order q, i.e. additional stages and fill, whereas the collocation families carry their matching-order continuous extension for free. In the measured work–precision curves the explicit methods are therefore competitive only in the low-accuracy regime ($\varepsilon \gtrsim 10^{-4}$), and Gauss stages dominate everywhere beyond it. [quantify crossover from work-precision CSVs after suite completion]

Remark on scope: smooth systems

The high-order convergence results above — like those of every spectral or high-order method — presuppose (piecewise-)smooth coefficients with any discontinuities aligned to grid nodes. For genuinely non-smooth systems (act-and-wait switching, interrupted cutting) *all* methods of any nominal order fall back to low observed order when the discontinuity falls inside a step; this is verified for the act-and-wait beam in Appendix . Two practical consequences: the sweet-spot recommendation of moderate order with many steps holds *a fortiori* (the sparsest variants lose nothing where high order cannot pay off), and the linear-complexity advantage of the solution-operator assembly is unaffected. A discontinuity-aware treatment (mesh alignment, step splitting) is outside the scope of this paper.

Discussion: practical guidance

The measurements condense into a short decision rule for practitioners.

1. **Default: Gauss stages, moderate order.** Choose $s \approx \max\bigl(2,\ \lceil \ln(1/\varepsilon)/6 \rceil\bigr)$ for a relative accuracy target ε (Equation 9); even $\varepsilon = 10^{-12}$ asks only for $s \approx 5$. Then pick p by one or two refinement

steps until the target is met — the cost is linear in p, so overshooting p is cheap, whereas overshooting s inflates the fill cubically.

2. **Large systems: cap the order earlier.** The per-step fill scales with d^2s^3; for FEM-scale models ($d \gtrsim 20$) the measured optima sit at $s = 2$–3 [confirm exact beam optima], and memory, not CPU, becomes the binding constraint for $s \gtrsim 5$.
3. **Avoid the spectral corner for engineering models.** The single-step (global collocation) mode is elegant for small, smooth, low-accuracy-run-count problems, but it abandons sparsity (nnz $\sim d^2s^3$ in one dense block) and loses its order advantage entirely for non-smooth coefficients (act-and-wait, interrupted cutting).
4. **Explicit steps only for cheap low-accuracy sweeps on non-stiff systems.** Their observed order is capped by the available dense output (≈ 3 for classical RK4), and on stiff high-dimensional systems they are unusable below a CFL-like resolution bound, where the A-stable Gauss stages converge from the coarsest grids.
5. **Non-smooth coefficients: align or refine.** If switching instants can be placed on grid nodes, full order is retained; if not, expect order collapse and prefer moderate order with more steps.

Conclusion

Once the stability analysis of time-periodic DDEs is written in multiplication-free banded form, the per-step integrator is a free design choice: Butcher-tableau residual blocks with continuous-extension delay interpolation raise the convergence order to that of the chosen scheme, up to Gauss super-convergence $2s$, while retaining $\mathcal{O}(p)$ complexity. The two classical method families are the endpoints of one continuum — piecewise-constant semi-discretization at maximal sparsity and global collocation at maximal per-step order — and neither endpoint is optimal: the CPU-time optimum sits at a problem-dependent sweet spot of moderate order and many steps, drifting only logarithmically with the accuracy target and towards lower order for larger system dimension and non-smooth coefficients. This unifies and clarifies, rather than reinvents; the actionable outputs are the sweet-spot guidance and the open-source SOSD.jl implementation.

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Declaration of conflicting interests

The author declares no conflict of interest.

Model parameters

Delayed Mathieu equation $(d = 2)$

$$\ddot{x} + a_1\dot{x} + \big(\delta + \varepsilon\cos(2\pi t/T)\big)x = b_0\, x(t-\tau) + \sin(4\pi t/T),$$

with $\delta = 3,\ \varepsilon = 0.2,\ b_0 = -0.15,\ a_1 = 0.1,\ \tau = T = 2\pi$. Converged dominant multiplier $\mu = 0.351417260953549$ (Table, two-resolution GL10 reference).

Seasonal scalar model $(d = 1)$ $\dot{y}(t) = -(a + b\cos t)\, y(t-\tau)$ with $a = b = 1$, $\tau = 2$, $T = 2\pi$; $\mu = 0.626021929290183$.

Turning with spindle-speed variation $(d = 2)$

$$\ddot{x} + \zeta\dot{x} + (1+k_w)x = k_w\, x\big(t-\tau(t)\big) + \cos(2\pi t), \quad \tau(t) = \frac{2\pi}{\Omega}\Big(1 + A_{\mathrm{SSV}}\sin\tfrac{2\pi t}{T}\Big),$$

with $k_w = 0.2,\ \zeta = 0.1,\ \Omega = 0.3,\ A_{\mathrm{SSV}} = 0.1,\ T = N_T\, 2\pi/\Omega,\ N_T = 10$ (so $T/\tau_{\max} \approx 9.1$); $\mu \approx 4.47617$.

FEM beam with delayed boundary feedback $(d = 28)$ Longitudinal rod of length $L = 100\,\mathrm{m}$, Young's modulus $E = 210\,\mathrm{GPa}$, cross-section $A = 10^{-4}\,\mathrm{m}^2$, density $\rho = 7800\,\mathrm{kg/m}^3$, stiffness-proportional damping $\eta = 0.01$, discretized by $N = 15$ linear elements with lumped mass, fixed at the first node ($d = 2(N-1) = 28$ states). Delayed displacement feedback from the first free node to the last node with gain $P\,E/(L/N)$, $P = 0.2$; wave speed $c = \sqrt{E/\rho}$, $T_{\mathrm{sp}} = L/c$, delay $\tau = 0.2\,T_{\mathrm{sp}}$, period $T = 0.4\,T_{\mathrm{sp}}$ ($\tau/T = 1/2$); constant unit force at the free end. In the act-and-wait configuration the feedback matrix is active only for $t < 0.8\,T$. [beam reference μ from final CSV]

Additional work–precision studies

Figures 5–7 repeat the three-panel work–precision study of Figure 3 for the remaining test systems. The qualitative picture is identical: linear time complexity for every embedded integrator, Gauss stages dominating beyond $\varepsilon \approx 10^{-4}$. On the stiff beam the explicit methods are unstable below a CFL-like resolution bound (errors up to 10^{43}), while the A-stable Gauss stages converge from the coarsest grids.

Non-smooth coefficients: act-and-wait beam

With act-and-wait switching, the beam's feedback matrix jumps at $t = 0.8\,T$. On grids where the switching instant is a node (p divisible by 5) the full order is retained; on off-grid meshes (powers of two) the observed order of every method collapses towards one–two near the switching-induced error floor (Figure 8), confirming the scope remark of Section .

Reproducibility

All computations: Julia 1.12.4 on Windows 11 (x86-64), BLAS restricted to a single thread, package versions pinned by the repository manifests

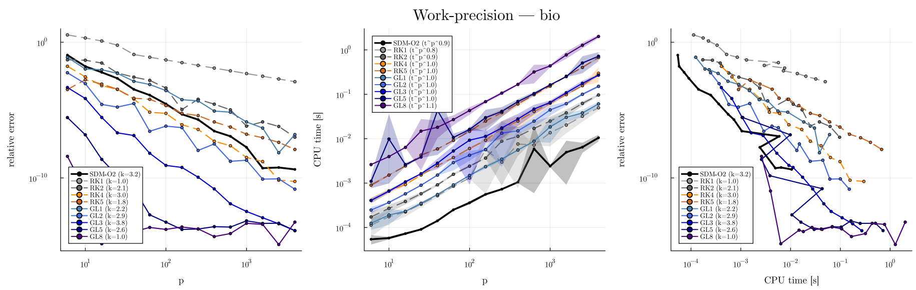

Figure 5. Work-precision — seasonal scalar model.

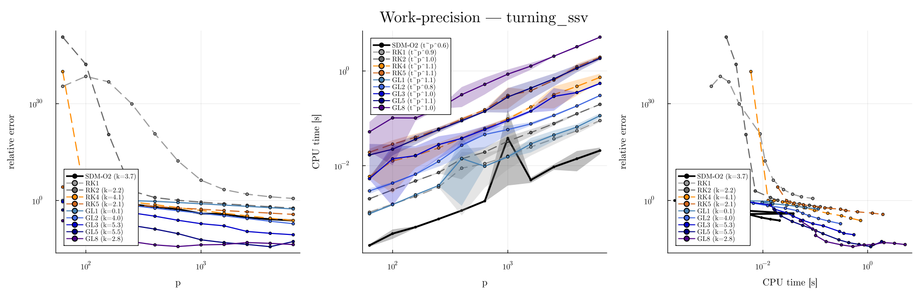

Figure 6. Work-precision — turning with spindle-speed variation.

Figure 7. Work-precision — FEM beam with delayed boundary feedback.

`pending: figures/nonsmooth_beam.pdf`

Figure 8. Act-and-wait beam: relative error versus p for GL1–GL3 on grid-aligned (solid) and off-grid (dashed) meshes.

(SOSD.jl v0.1.1 with KrylovKit, LinearMaps, RungeKutta, StaticArrays; baseline SemiDiscretizationMethod.jl v0.4.2). Timings are means over repeated warm-started runs with standard deviations shown as bands; start vectors of all Krylov iterations are deterministic. Reference multipliers are accepted only when two resolutions of a GL(10) (small systems) or lazy GL(5) (beam) computation agree to 10^{-10} relative, and the reference method is cross-validated against SemiDiscretizationMethod.jl on every system. Every figure regenerates via `julia --project=benchmark benchmark/run_all.jl` followed by `benchmark/make_figures.jl`. [repository URL + commit hash at submission]

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